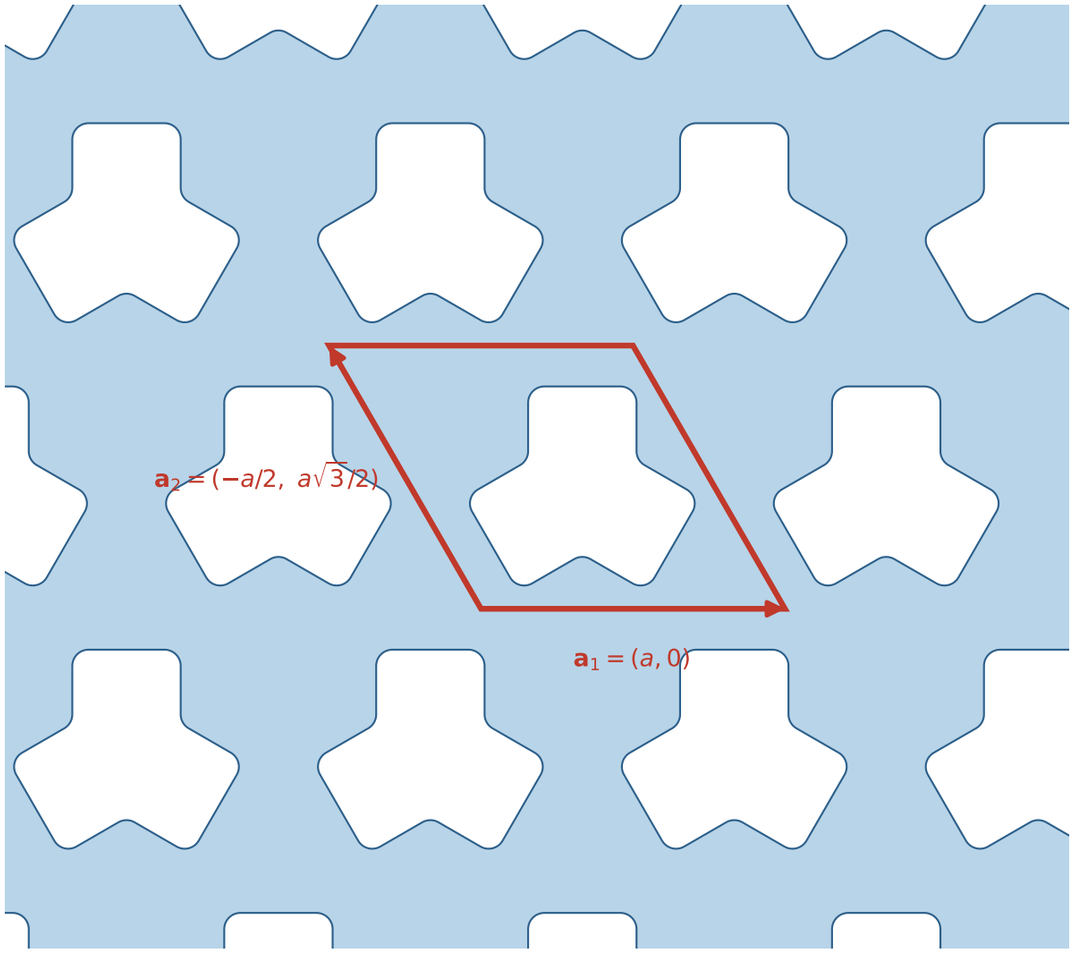
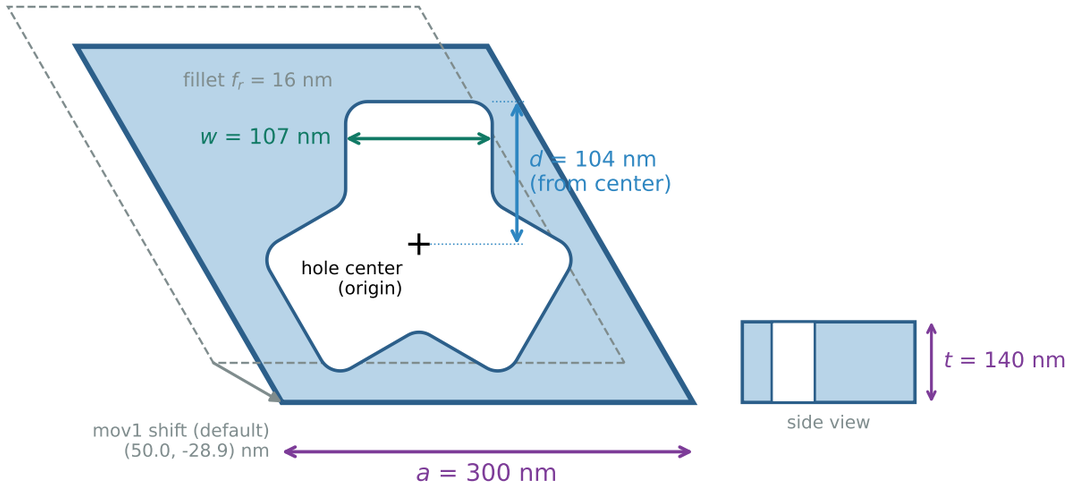
Diamond snowflake phononic crystal -- d x w sweep (diamond_dw_sweep)
unit_cell_3.mph | a=300, f_r=16, t=140 nm | E=1050 GPa, nu=0.07, rho=3500 kg/m^3 | N_mode=10, n_seg=7 (21 k-pts)
panels A-C: nominal d=104, w=107 nm

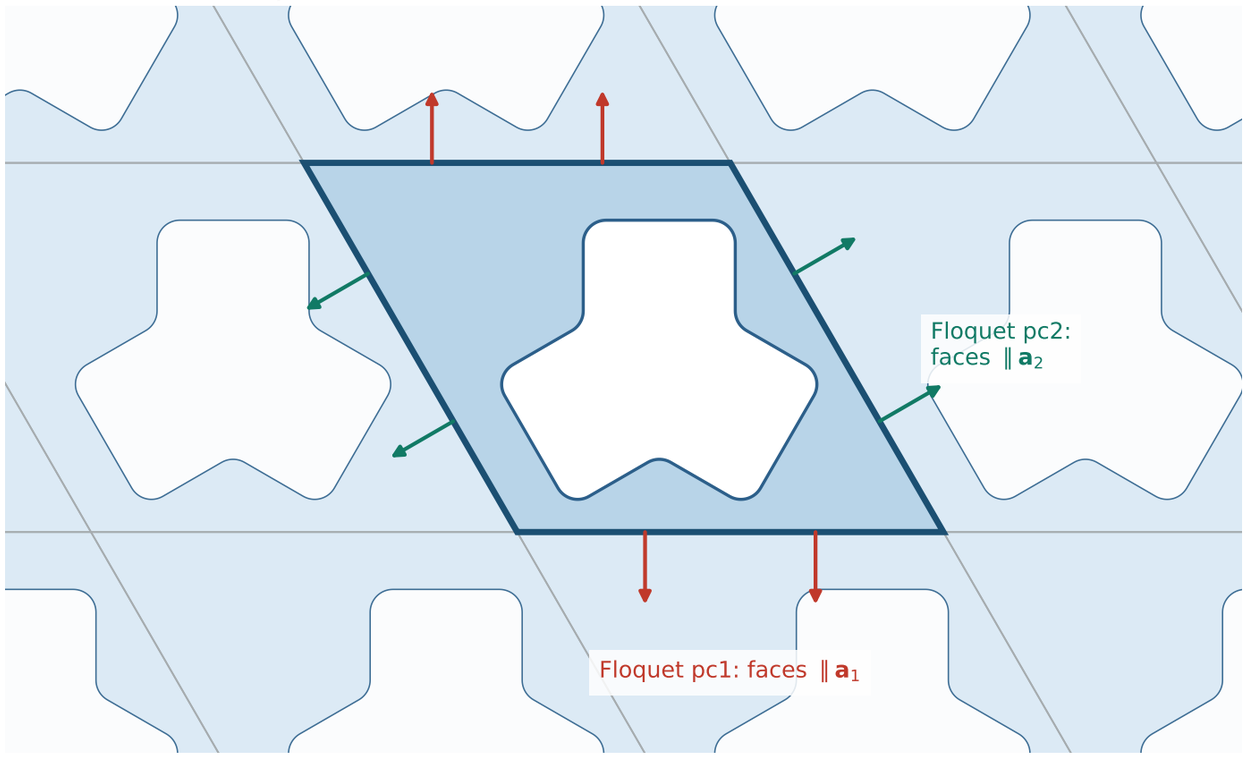
(A) Triangular lattice -- rhombus primitive cell (as in pol1)



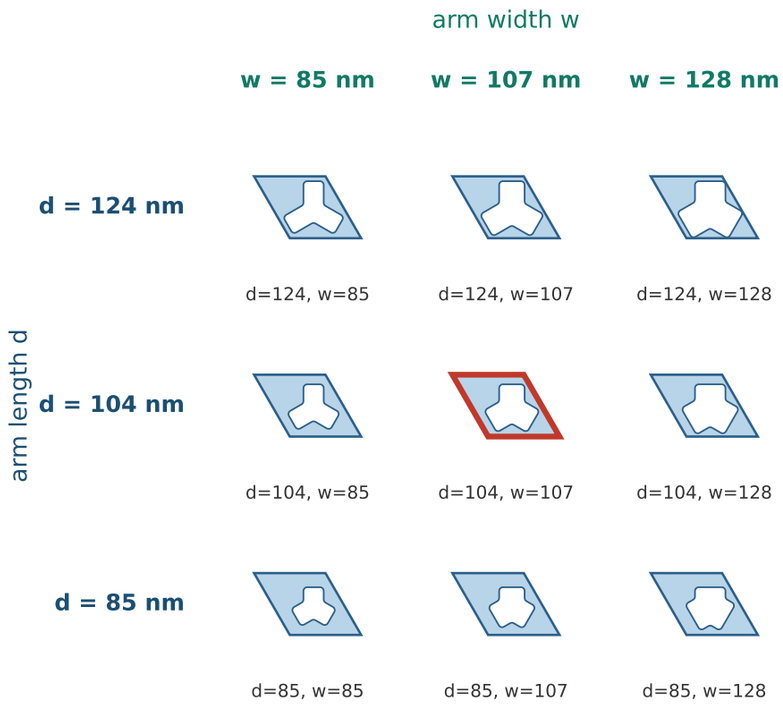
(B) Unit cell (top view): hole at lower-triangle centroid, 3 arms of length d from the center at 90/210/330 deg



(C) COMSOL cell + Floquet BCs (pc1, pc2)
$\mathbf{k} = (k_{Fx}, k_{Fy}, 0)$ Cartesian; z=0 midplane: sym1 / as1; top face free



(D) 3x3 sweep grid (cells to scale, default mov1 shift)



red = nominal (diamond_mech_v1)